Born with it, or Breathed it in?

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Dataset Overview

Data Types

Primarily made up of integers, as the dataset contains patient symptoms ranked by severity on a 1-7 scale.

```
table(CPDS$Level)

##
##    High     Low Medium
##     365     303     332
```

Sample Size

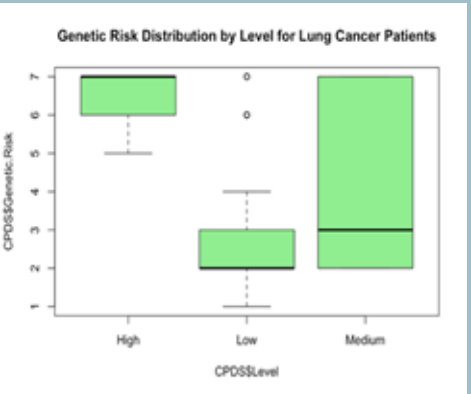
The kaggle dataset contains 1,000 patient records. All recorded patients have lung disease.

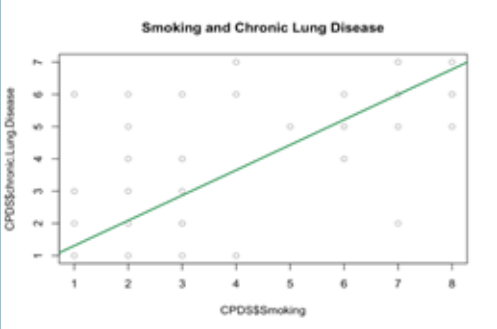
```
summary(CPDS)

##      index        Patient.Id            Age            Gender
##  Min.   :  0.0   Length:1000        Min.   :14.00   Min.   :1.000
##  1st Qu.:249.8   Class :character   1st Qu.:27.75   1st Qu.:1.000
##  Median :499.5   Mode  :character   Median :36.00   Median :1.000
##  Mean   :499.5                      Mean   :37.17   Mean   :1.402
##  3rd Qu.:749.2                      3rd Qu.:45.00   3rd Qu.:2.000
##  Max.   :999.0                      Max.   :73.00   Max.   :2.000
##  Air.Pollution   Alcohol.use     Dust.Allergy   OccuPational.Hazards
##  Min.   :1.00   Min.   :1.000   Min.   :1.000   Min.   :1.00
##  1st Qu.:2.00   1st Qu.:2.000   1st Qu.:4.000   1st Qu.:3.00
##  Median :3.00   Median :5.000   Median :6.000   Median :5.00
##  Mean   :3.84   Mean   :4.563   Mean   :5.165   Mean   :4.84
##  3rd Qu.:6.00   3rd Qu.:7.000   3rd Qu.:7.000   3rd Qu.:7.00
##  Max.   :8.00   Max.   :8.000   Max.   :8.000   Max.   :8.00
##   Genetic.Risk  chronic.Lung.Disease Balanced.Diet      Obesity
##  Min.   :1.00   Min.   :1.00         Min.   :1.000   Min.   :1.000
##  1st Qu.:2.00   1st Qu.:3.00         1st Qu.:2.000   1st Qu.:3.000
##  Median :5.00   Median :4.00         Median :4.000   Median :4.000
##  Mean   :4.58   Mean   :4.38         Mean   :4.491   Mean   :4.465
##  3rd Qu.:7.00   3rd Qu.:6.00         3rd Qu.:7.000   3rd Qu.:7.000
##  Max.   :7.00   Max.   :7.00         Max.   :7.000   Max.   :7.000
##     Smoking       Passive.Smoker    Chest.Pain    Coughing.of.Blood
##  Min.   :1.000   Min.   :1.000   Min.   :1.000   Min.   :1.000
##  1st Qu.:2.000   1st Qu.:2.000   1st Qu.:2.000   1st Qu.:3.000
```

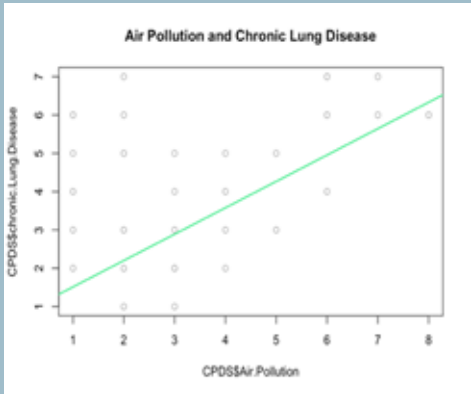
Descriptive Analysis

Correlation Coefficients

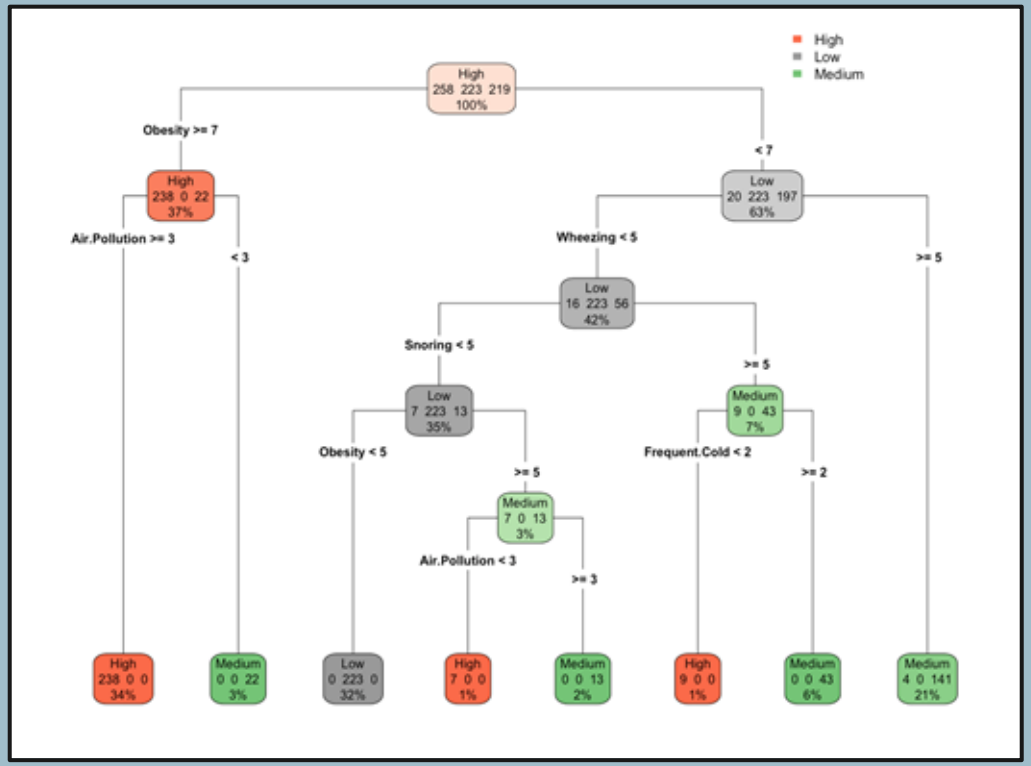
Occupational Hazards	0.8582839 - Highest Correlation
Genetic Risk	0.8362308
Air Pollution	0.6267009
Balanced Diet	0.6226321
Dust Allergy	0.6195559
Smoking	0.5785853







Predictive Analysis



Optimization

A biomedical company, PulmoTech, manufactures two types of respiratory devices for patients with chronic lung diseases: portable oxygen concentrators and advanced nebulizers. These devices are crucial in managing conditions like COPD and severe asthma.

Each oxygen concentrator provides a profit of $250, while each nebulizer provides a profit of $180.

```
optimum$solution
```

```
## [1]  0 40
```

```
optimum$objval
```

```
## [1] 7200
```

```
optimum$duals
```

```
## [1]  60   0 -50   0
```

Any Questions?

Thank you!